VERIFIED CONDITIONAL CONSTRUCTIONS

Two routes from H00 to RH

A mathematical and reproducibility record for the v0.2.3 native checkpoint.

The project supplies two forward constructions from the same semantic zero-cancellation hypothesis to Mathlib's literal Riemann Hypothesis. Each is paired with the retained converse. The native run is reported as compiled and axiom-audited; GitHub CI is a new, separately tracked verification target.

$$H_{00} \Leftrightarrow \text{RiemannHypothesis}$$

The exact Lean endpoints

All four declarations live in the namespace `RH_HilbertLimit.TwoRoutes`.

```
nativeRH_of_H00_route_one :
 H00 → _root_.RiemannHypothesis

H00_iff_nativeRH_route_one :
 H00 ↔ _root_.RiemannHypothesis

nativeRH_of_H00_route_two :
 H00 → _root_.RiemannHypothesis

H00_iff_nativeRH_route_two :
 H00 ↔ _root_.RiemannHypothesis
```

Two forward constructions, shared foundations

Route 1: zero cancellation gives constant Bernstein-row variation, then subexponential growth and the retained actual spectral closure.

Route 2: row positivity constructs a probability measure with the actual moments. Its analytic resolvent identifies the actual logarithmic derivative and excludes nonreal zeros.

The second forward proof does not call Route 1's RH endpoint. It does use shared finite positivity lemmas. Both equivalence declarations use the same retained RH-to-H00 converse.

Scope boundary. H00 is assumed, not established. No finite accepted certificate or unconditional proof of RH is claimed. The original longer Widder/angular-PDE chain remains a separate, unfinished formalization.

Source anchors · RH_HilbertLimit/TwoRoutes/RouteOne.lean; RouteTwo.lean; native PASS companion checkpoint.

01 / EXACT PREMISE

Finite data, universal validity

The zero-budget pair is finite. Its admissibility is an all-order proposition.

Let Θ denote `AWPIRH.theta`, the project's actual entire theta function. Its logarithmic derivative and normalized real moments are the unchanged objects used by both routes:

$$R(z) = \Theta'(z) / \Theta(z)$$
$$m_n = \mathrm{Re}[(-1)^n R^{(n)}(0) / n!]$$
$$\kappa = m_0 > 0, \quad u_n = m_n / \kappa^{n+1}, \quad u_0 = 1$$

Reality of the complex moments is supplied by the retained Schwarz-reflection result. Positivity of κ and the degree-zero normalization are unconditional ingredients, not additional assumptions of either endpoint.

Bernstein rows and cancellation

$$w_{N,j} = \mathrm{binom}(N,j) \sum_{r=0}^{N-j} (-1)^r \mathrm{binom}(N-j,r)\, u_{j+r}$$
$$V_N = \sum_{j=0}^{N} |w_{N,j}|$$
$$C_n = (n+1)(V_{n+1} - V_n) \geq 0$$

Here C_n is `rowCancellation`. The displayed variation identity is proved by finite Bernstein coarsening. The named candidate is exactly the natural-number pair with cutoff 0 and coefficient 0.

```
def H00 : Prop := zeroBudgetCandidate.Admissible
```

For a budget W, admissibility means every n at or beyond W.cutoff satisfies $C_n \leq$ W.coefficient $\cdot V_n$. Thus the candidate (0,0) requires all C_n to vanish:

$$H_{00} \Leftrightarrow (\forall n,\ C_n = 0)$$
$$\Leftrightarrow (\forall N,\ V_N = 1)$$
$$\Leftrightarrow (\forall N, \forall j \leq N,\ w_{N,j} \geq 0)$$

These are semantic equivalences. A finite list of successful rows does not establish the universal condition. The required propagation or checker argument remains open.

Source anchors · ActualXi.lean; Bernstein.lean; Cancellation.lean; ActualCancellationWitness.lean; ZeroBudgetCandidate.lean; TwoRoutes/RouteOne.lean.

02 / ROUTE 1

Cancellation to spectral closure

The forward chain begins with exact finite identities and ends at the retained actual spectral theorem.

1. Every variation increment vanishes

H00 gives $C_n = 0$ at every index. Since n+1 is positive, the cancellation identity forces $V_{n+1} = V_n$. Induction from the unconditional $V_0 = 1$ gives $V_n = 1$ for all n.

```
actual_variation_one_of_H00
actual_weights_nonnegative_of_H00
```

The row-mass identity and unit total variation also force every Bernstein weight to be nonnegative. This finite consequence is the common input used by Route 2.

2. Unit variation is subexponential

$$\forall \varepsilon > 0, \quad \forall n \geq 0, \quad V_n = 1 \leq \exp(\varepsilon n)$$

The source supplies the constant 1 in the definition of `Subexponential actualVariation`. There is no extra growth premise and no numerical asymptotic test.

```
actual_subexponential_of_H00
```

3. Apply the actual spectral equivalence

The retained theorem `canonical_actual_subexponential_iff_nativeRH` identifies subexponential actual variation with literal Mathlib RH. The route invokes its forward direction on the bound just derived.

```
theorem nativeRH_of_H00_route_one (h : H00) :
 _root_.RiemannHypothesis := by
 exact canonical_actual_subexponential_iff_nativeRH.mp
 (actual_subexponential_of_H00 h)
```

The converse is shared

The retained RH-to-zero-cancellation construction supplies `H00_of_nativeRH`. Pairing it with this forward proof yields `H00_iff_nativeRH_route_one`. Route 2 uses the same converse only when forming its own equivalence.

Source anchors · TwoRoutes/RouteOne.lean; ZeroBudgetCandidate.lean; ActualCancellationWitness.lean; retained actual spectral closure imported by these modules.

03 / ROUTE 2: MEASURE CONSTRUCTION

Positive rows become a measure

The measure is constructed from H00; it is not an additional existence hypothesis.

1. Positive finite averages

For positive Bernstein rows, the finite averages have total mass one. Writing $q_{N,j} = j/N$ for $N > 0$, define the corresponding finite quadrature on [0,1]:

$$A_N(f) = \sum_{j=0}^{N} w_{N,j}\, f(q_{N,j})$$
$$A_N(1) = 1, \quad A_N(f) \geq 0 \text{ if } f \geq 0$$
$$|A_N(f)| \leq \|f\|_\infty$$

The degree-zero definition is handled in the source; it does not affect the limit at infinity. The exact Bernstein monomial identity gives $A_N(q \mapsto q^k) \to u_k$ for each k.

`bernsteinAverage_monomial_tendsto`

2. A positive functional on continuous functions

Compact bounds permit a hyperfilter limit of the finite averages for each continuous f. Linearity, positivity and normalization pass to the limit. The result is a positive linear functional L on $C([0,1], \mathbb{R})$, with $L(1) = 1$.

This is a classical, noncomputable construction. It is compatible with the stated axiom allowlist; it does not turn H00 into a terminating decision procedure.

3. Riesz representation and exact moments

Mathlib's real Riesz representation theorem constructs the representing measure σ. Normalization gives total mass one. The independently proved monomial limit identifies all its moments:

$$\sigma([0,1]) = 1, \quad \sigma \geq 0$$
$$\int_{[0,1]} q^k \, d\sigma(q) = u_k \quad \text{for every } k \in \mathbb{N}$$

```
actualHausdorffMeasure (h : H00)
actualHausdorffMeasure_isProbabilityMeasure
actualHausdorffMeasure_moment
```

The actual measure is obtained by instantiating the generic construction with u = actualNormalizedMoment and the positivity already derived from H00. No output of the first RH closure is used.

Source anchors · TwoRoutes/BernsteinMomentLimit.lean; HausdorffMeasure.lean; shared finite positivity lemmas in RouteOne.lean and PositiveMomentBridge.lean.

04 / ROUTE 2: ANALYTIC CLOSURE

The resolvent excludes nonreal zeros

The continuation step uses an analytic defect across possible zeros of Θ.

1. Form the positive resolvent

$$S(z) = \kappa \int_{[0,1]} 1 / (1 + \kappa q z)\, d\sigma(q)$$
$$\operatorname{Im} S(z) \le 0 \quad \text{whenever } \operatorname{Im} z > 0$$

Compact support gives analyticity near zero and throughout the upper half-plane. The kernel has nonpositive imaginary part there, and integration preserves this sign.

2. Identify the actual germ

$$S^{(n)}(0) = (-1)^n\, n!\, \kappa^{n+1}\, u_n = R^{(n)}(0)$$

Both functions are analytic near zero, where Θ(0) ≠ 0. Equality of all derivatives identifies their convergent Taylor germs. The source uses analytic power series, rather than merely comparing formal coefficients.

`actual_logDerivative_resolvent_germ`

3. Continue the analytic defect

$$D(z) = \Theta'(z) - S(z)\Theta(z)$$

D is analytic on the connected upper half-plane and vanishes on an open portion supplied by the origin germ. The identity theorem gives D = 0 there. This argument does not assume that Θ is zero-free in order to continue Θ′/Θ.

4. Exclude nonreal zeros and conclude RH

At points where Θ is nonzero, D = 0 gives R = S and hence Im R ≤ 0. A zero of multiplicity m in the upper half-plane would create the local logarithmic-derivative pole m/(z-α). Probing immediately below that zero forces a positive imaginary part, contradicting the sign bound. Conjugation symmetry excludes lower-half-plane zeros, and the retained actual-theta zero bridge concludes literal RH.

`nativeRH_of_H00_via_measure_resolvent`
`nativeRH_of_H00_route_two`

All analytic and measure obligations in this compressed route are supplied internally. The original Widder/angular-PDE route is not certified by this replacement.

Source anchors · TwoRoutes/MeasureResolvent.lean; ResolventAnalytic.lean; ActualResolventContinuation.lean; ActualHerglotzClosure.lean; RouteTwo.lean.

05 / EVIDENCE AND REPRODUCTION

What is verified, and where

Native evidence is preserved. GitHub CI is a separate target with its own run artifacts.

Native checkpoint: user-reported PASS

The supplied console excerpt reports PROJECT_BUILD=PASS and RUN=PASS: 3 modules compiled and 115 reused. The preserved plan contains 117 source modules plus the final audit source, with 641 selected declarations. Inspection of that plan confirms coverage of both endpoints and both equivalences.

The checkpoint records that native logs and receipts were **not independently read**. Exact per-declaration axiom lists are absent from the supplied excerpt. Native success is therefore recorded as reported by the user, supported by audit-plan and runner inspection.

New target: GitHub CI

The CI setup pins Lean 4.33.0 and mathlib's v4.33.0 commit. The native mathlib commit was not supplied; this is a new reproducible CI baseline. GitHub Actions builds the preserved sources, runs the final axiom audit and retains logs. Missing declarations, malformed output and unexpected axioms must fail the audit.

```
Lean: leanprover/lean4:v4.33.0
Mathlib: db584cd6d46c92f209a44c0f1c829460d327499d
Allowed: propext, Classical.choice, Quot.sound
```

Publication status: GitHub CI is newly configured. This report does not assert a CI PASS; consult the workflow run associated with the repository commit. Native receipts are not substituted for a GitHub CI result.

Repository: randogelist/Hilber-Vector-Resolvent-Theory
Live GitHub Actions runs and audit artifacts

Preserved package identity

```
RH_H00_TWO_ROUTES_LEAN_v0_2_3_construction.zip
SHA-256:
578cc8ad072c79dc7838fcf65a27b9eed
707fcc98e0fdf6a4b3601aa20e8c51a
```

Open obligations

H00 admissibility and all-order propagation; a terminating finite checker with soundness and completeness; an accepted finite witness; the accepted-witness/RH equivalence; and the longer Widder/angular-PDE chain. Unconditional RH is not established.

The native package stays unchanged; its preparation-status headers predate the newer PASS checkpoint. The exact anchor ZETA_HYPERBOLIC_ANCHOR_5 is preserved. No new Init record or local Lean/Lake execution was used to prepare this report and setup.

Source anchors · RH_H00_TWO_ROUTES_LEAN_v0_2_3_NATIVE_PASS_CHECKPOINT.json; TwoRoutesAudit.lean; audit-targets.tsv; compile-order.txt; pinned CI configuration.